Customer Behavior Analysis: Project N2 Documentation

“e-commerce customer dataset”

Prepared by: Narges Sabbagh

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1. Introduction

Project Background

Customer data is one of the most valuable assets for any e-commerce business. By analyzing purchasing behavior, companies can identify their most valuable customers, recognize users who are likely to stop purchasing, evaluate marketing strategies, and make informed business decisions.

The objective of this project is to analyze customer behavior using Python and transform raw business data into practical insights that support managerial decision-making.

Rather than focusing only on statistical analysis, this project emphasizes business interpretation and actionable recommendations.

Business Objectives

The analysis aims to answer the following business questions:

- Who are the company's most valuable customers?
- Which customers are at risk of churn?
- Which cities are the best candidates for future marketing campaigns?
- Are discount campaigns actually effective?
- What practical actions can increase future sales?

2. Dataset Description

The dataset contains information about 60 customers and 17 different attributes describing purchasing behavior and customer profiles.

The analysis uses the cleaned dataset produced in the previous project. Before beginning the business analysis, the dataset was validated to ensure that there were no missing values, duplicated records, or incorrect data types.

Main features:

Feature	Description
customer_id	Unique customer identifier
city	Customer city
membership_tier	Membership level
purchase_count	Number of completed purchases
avg_order_value	Average value per order
total_spending	Total customer spending
last_purchase_days	Days since last purchase

returned_items	Number of returned products
satisfaction_score	Customer satisfaction score
discount_used	Whether discounts were used
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3. Business Analysis

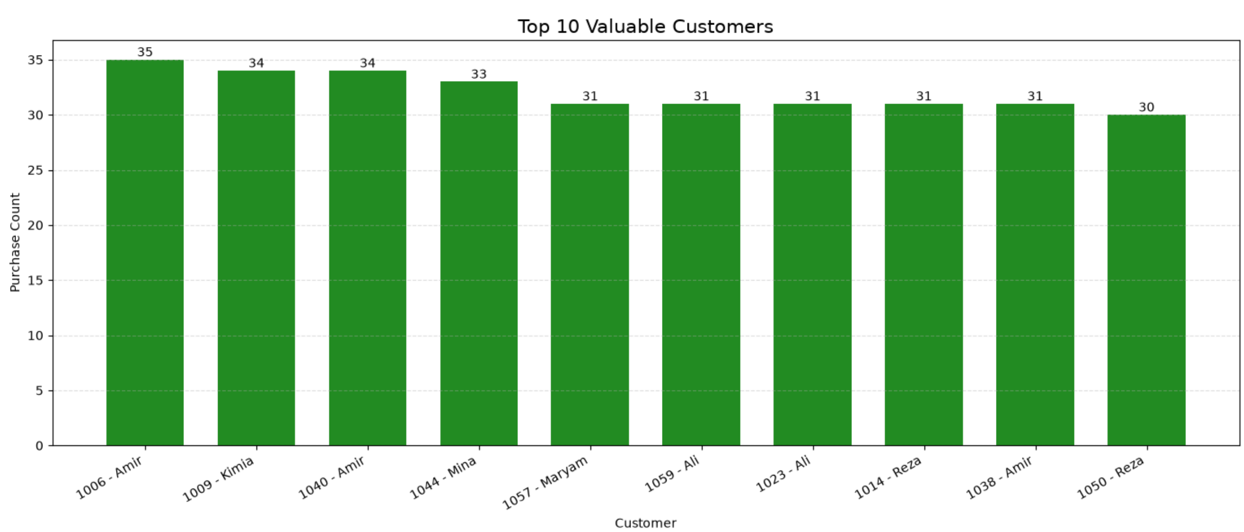
Who are the most valuable customers?

➢ Analysis Approach:

To identify valuable customers, two business indicators were selected:

- Purchase Count
- Total Spending

Customers were sorted primarily by purchase frequency and secondarily by total spending. The top ten customers were selected for further analysis.



➢ Observation:

The analysis shows that customers with the highest purchase frequency also contribute a significant share of the company's revenue. These customers represent the company's most loyal customer segment.

- Business Impact:

The company should focus on retaining these customers by offering loyalty rewards, personalized promotions, and exclusive benefits. Losing a high-value customer would have a greater financial impact than losing an occasional buyer.

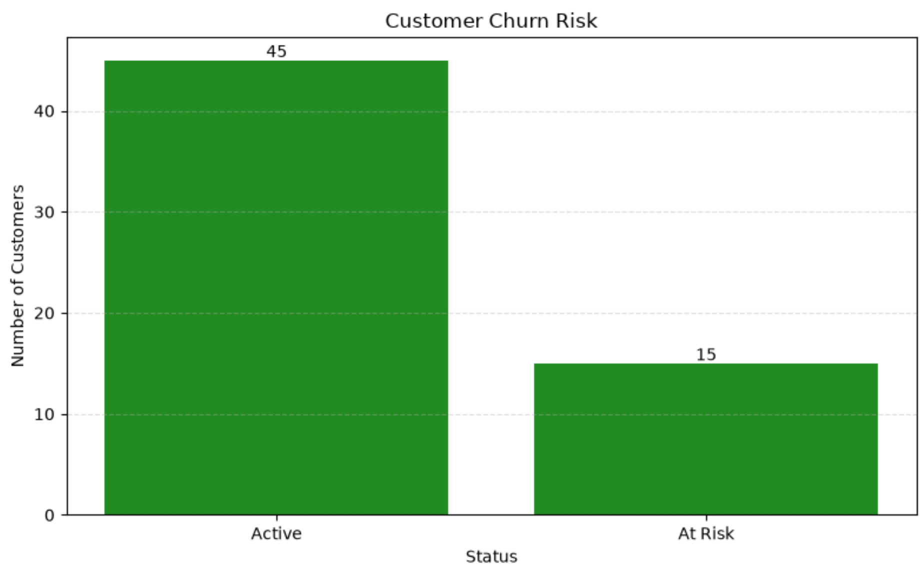
Which customers are at the risk of churning?

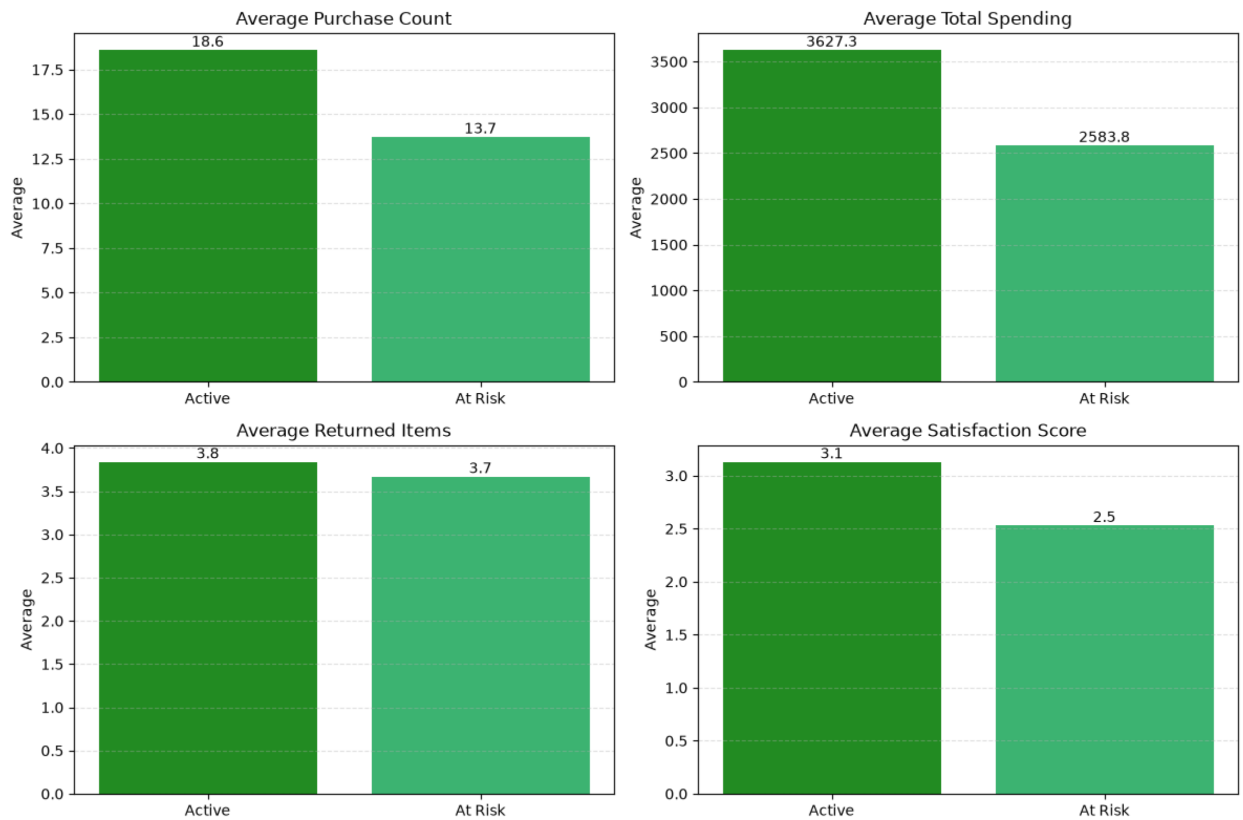
- Analysis Approach:

Customer churn risk was evaluated using three business indicators:

- Last Purchase Days
- Returned Items
- Satisfaction Score

Customers with long periods of inactivity were classified as potentially at risk of churn.





➢ Observation:

At-risk customers generally purchase less frequently, spend less money, return more products, and report lower satisfaction scores than active customers.

These patterns indicate declining engagement before customers eventually stop purchasing.

➢ Business Impact:

Early identification of these customers allows the company to launch retention campaigns such as reminder emails, personalized discounts, or customer support follow-ups before losing them completely.

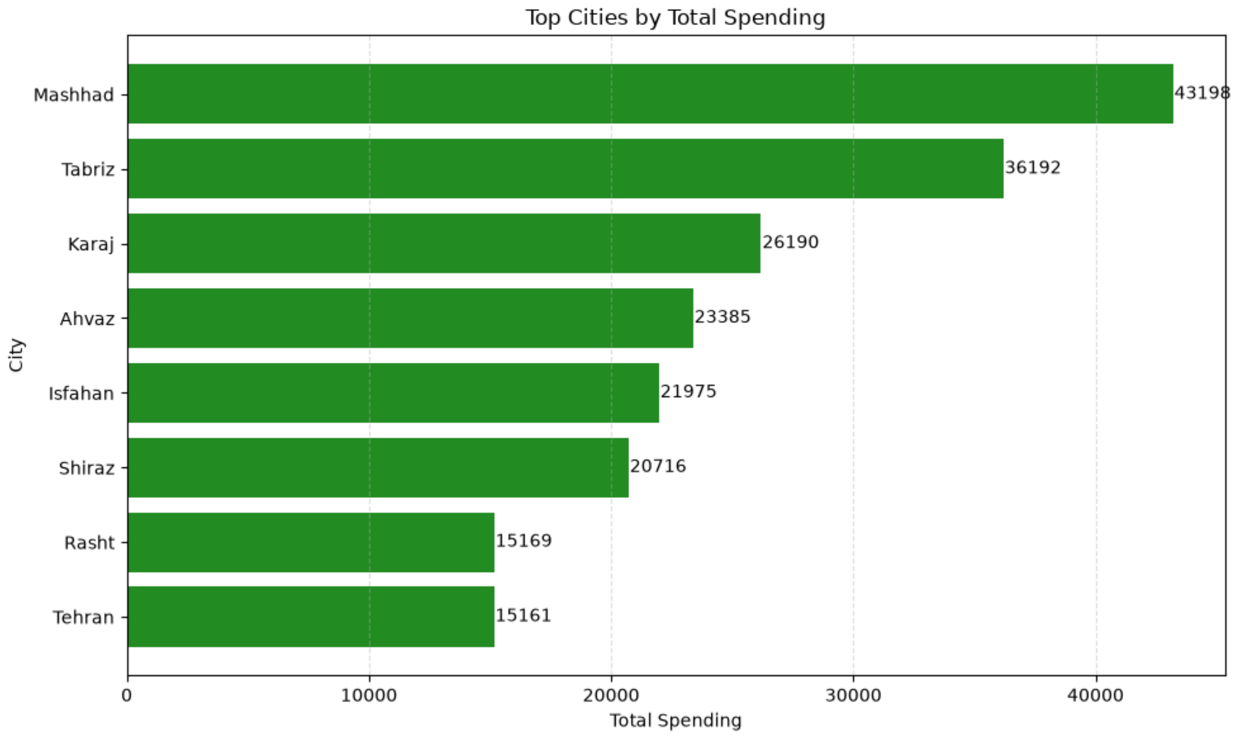
Which cities are best for marketing?

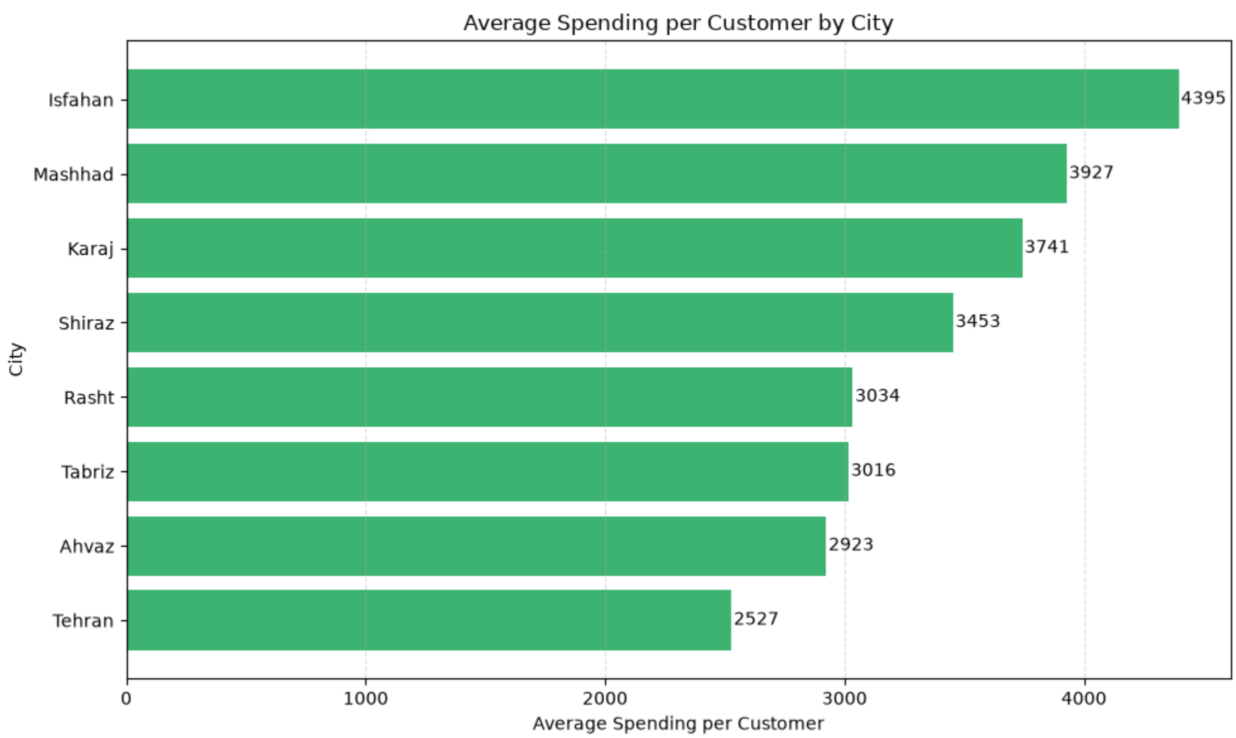
➢ Analysis Approach:

Cities were evaluated using two indicators:

- Number of Unique Customers
- Total Spending

Using both metrics provides a balanced view of market size and revenue potential.





- Observation:

Some cities contain a larger customer base, while others generate higher overall revenue. The strongest markets are those that perform well in both indicators.

➢ Business Impact:

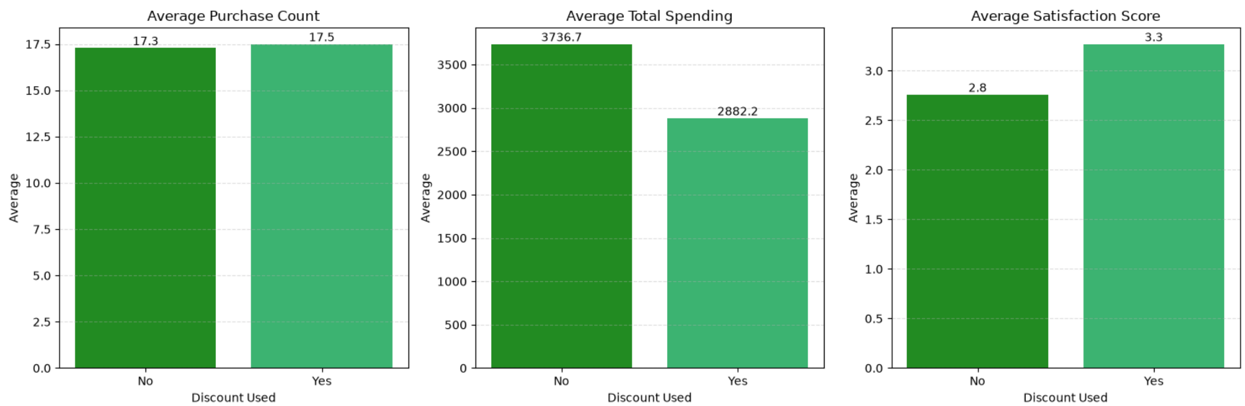
Marketing resources should be allocated based on customer concentration and purchasing power rather than population size alone. This approach increases the efficiency of advertising investments.

Are discounts really effective?

➢ Analysis Approach:

Customers who used discounts were compared with customers who did not using three performance indicators:

- Purchase Count
- Total Spending
- Satisfaction Score



➢ Observation:

Customers who used discounts purchased slightly more often and reported higher satisfaction scores. However, their average total spending was lower than customers who did not use discounts.

This suggests that discounts improve engagement but do not necessarily maximize revenue.

➢ Business Impact:

Discount campaigns should be targeted rather than offered to all customers. Personalized promotions can increase customer satisfaction while protecting overall profitability.

4. Business Recommendations

Based on the analysis, the following recommendations are proposed.

Recommendation 1: Strengthen Customer Loyalty

Create a loyalty program for high-value customers by providing exclusive rewards, personalized offers, and early access to promotions.

Expected Business Benefit:

- Higher customer retention
- Increased customer lifetime value
- More stable long-term revenue

Recommendation 2: Reduce Customer Churn

Identify inactive customers early and launch personalized retention campaigns before they stop purchasing completely.

Possible actions include reminder emails, special discounts, or customer support follow-ups.

Expected Business Benefit:

- Lower churn rate
- Better customer engagement
- Reduced revenue loss

Recommendation 3: Optimize Marketing Investment

Allocate marketing budgets toward cities with both a large customer base and high total spending instead of distributing advertisements equally across all regions.

Expected Business Benefit:

- Higher return on marketing investment (ROI)
- More efficient customer acquisition
- Better allocation of advertising budgets

5. Conclusion

This project demonstrates how customer transaction data can be transformed into meaningful business insights using Python and data analysis techniques.

The analysis successfully identified valuable customers, detected customers at risk of churn, evaluated city-level marketing opportunities, and assessed the effectiveness of discount campaigns.

These findings support data-driven decision-making and provide practical recommendations for improving customer retention, optimizing marketing strategies, and increasing long-term business performance.